Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

A retail company stores its business data across flat files (Excel/CSV) and a MySQL database. As a Data Analyst, the task is to connect Power BI Desktop to these data sources, import the required data, explore the Power BI interface, and create basic visuals to analyze sales by product, store, customer, and time. Management wants a centralized reporting system that consolidates data from multiple sources so that business performance can be monitored effectively.

Methodology / Steps Followed

Step 1: Data Collection

- Created a new database named retail_db in MySQL Workbench and set up a Product table with ProductID, ProductName, Category, and SubCategory columns.
- Populated the Product table using an SQL script based on the Kaggle Sample Superstore dataset.
- Verified the data load using SELECT COUNT(*) FROM Product;, confirming 1862 product records were inserted successfully with 0 duplicates and 0 warnings.
- Obtained Sales, Customer, Store, and Time data separately as flat files (Excel/CSV) for direct import into Power BI.

Step 2: Data Import

- Power BI Desktop was opened, and a new report file (Retail_dataset_exp1) was created from the Home screen.
- The Get Data option was used to connect to the MySQL database by specifying the server address (127.0.0.1:3306) and the database name (retail_db).
- The Sales, Customer, Store, and Time datasets stored as Excel/CSV flat files were imported using the same Get Data workflow.

Step 3: Data Cleaning and Transformation

- Opened all imported tables in the Power Query Editor.
- Assigned correct data types to each column, such as Whole Number for Row ID, Text for Order ID, and Date for Order Date/Ship Date.
- Promoted column headers from the first row and checked column names for consistency.
- Checked the Valid/Error/Empty indicators on each column to confirm the data was clean before loading.

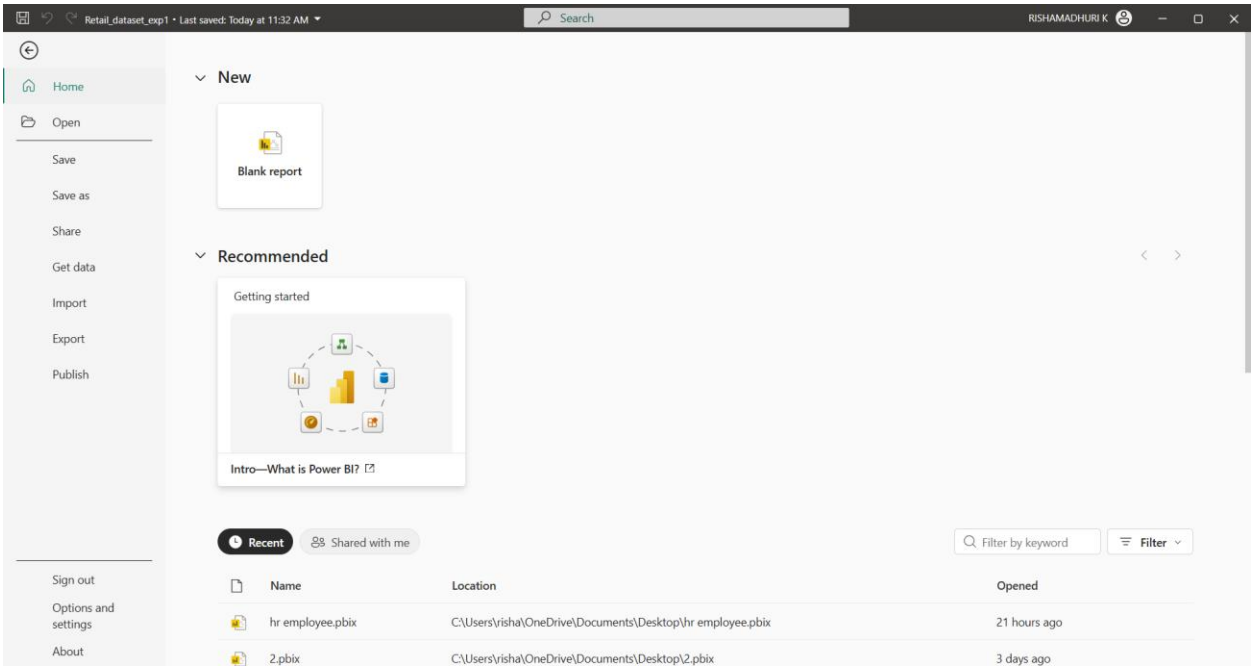
Step 4: Data Modeling

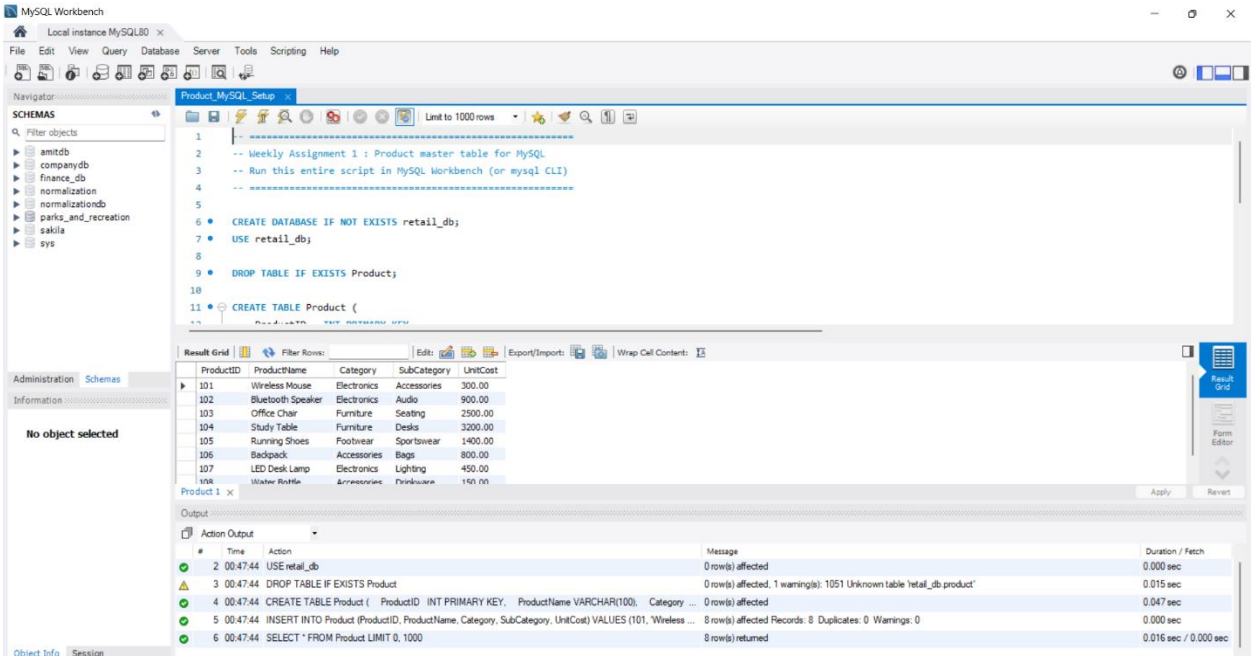
- Created relationships linking the Sales_dataset (fact table) to the Customer_Data, retail_db product, Store_Data, and Time_Data tables.
- Verified each relationship was one-to-many (one Customer/Product/Store/Date to many Sales rows), forming a star-schema data model.

Step 5: Dashboard Development

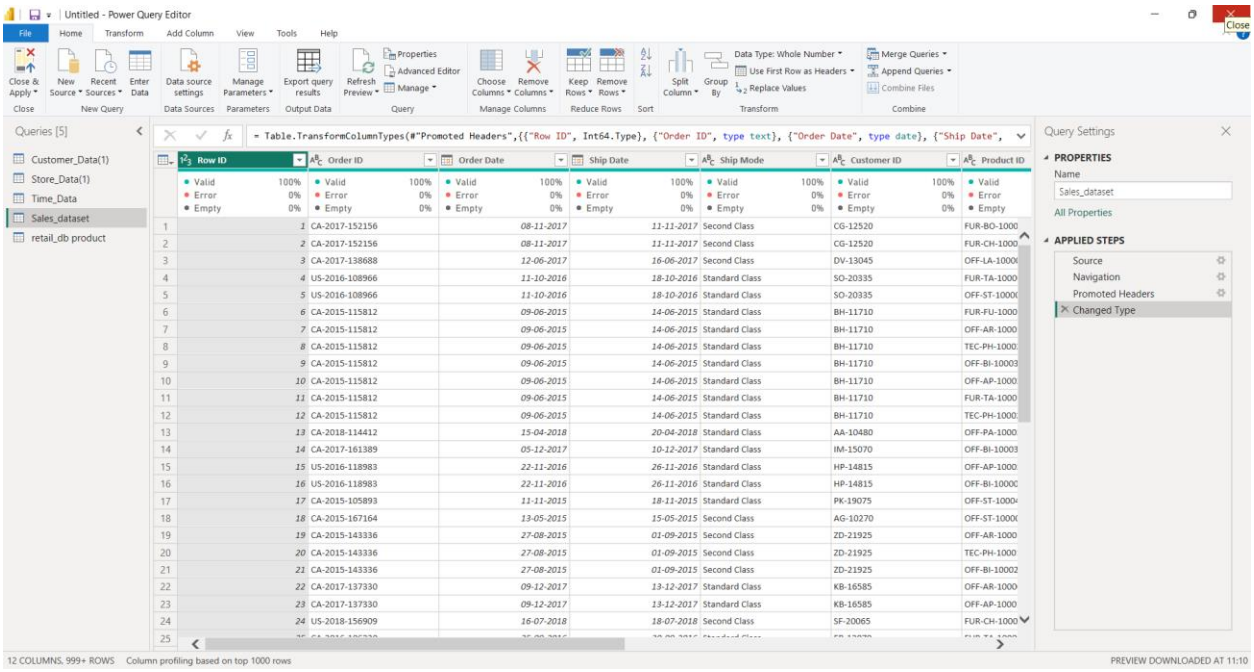
- Created DAX measures on the Sales_dataset table — Total Sales, Total Profit, Total Orders, Total Quantity, and Total Customers.
- Added ratio measures — Profit Margin %, Revenue per Order, and Units per Order.
- Built KPI cards to display these measures at a glance.
- Added charts and visuals — a map (Sales & Profit by State), a scatter chart (Discount vs. Profitability), a line chart (Monthly Sales Trend), and a table (Regional Profitability by Category).
- Added filters and slicers to allow drill-down by category, region, segment, and time.

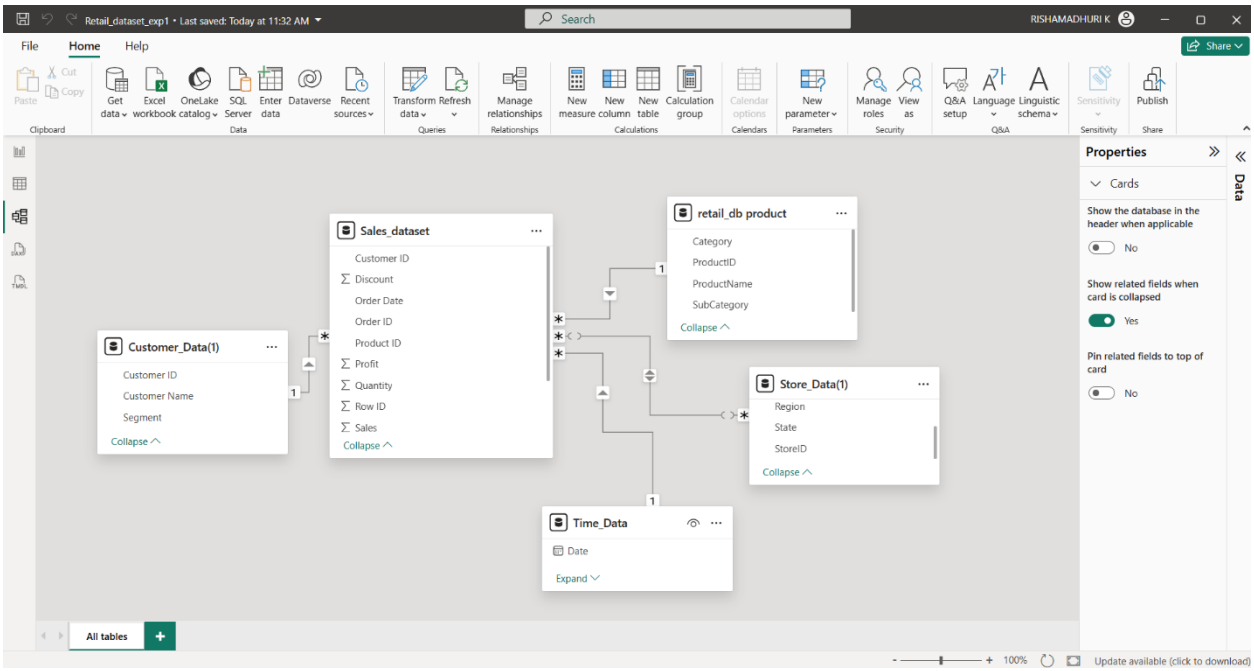
Output

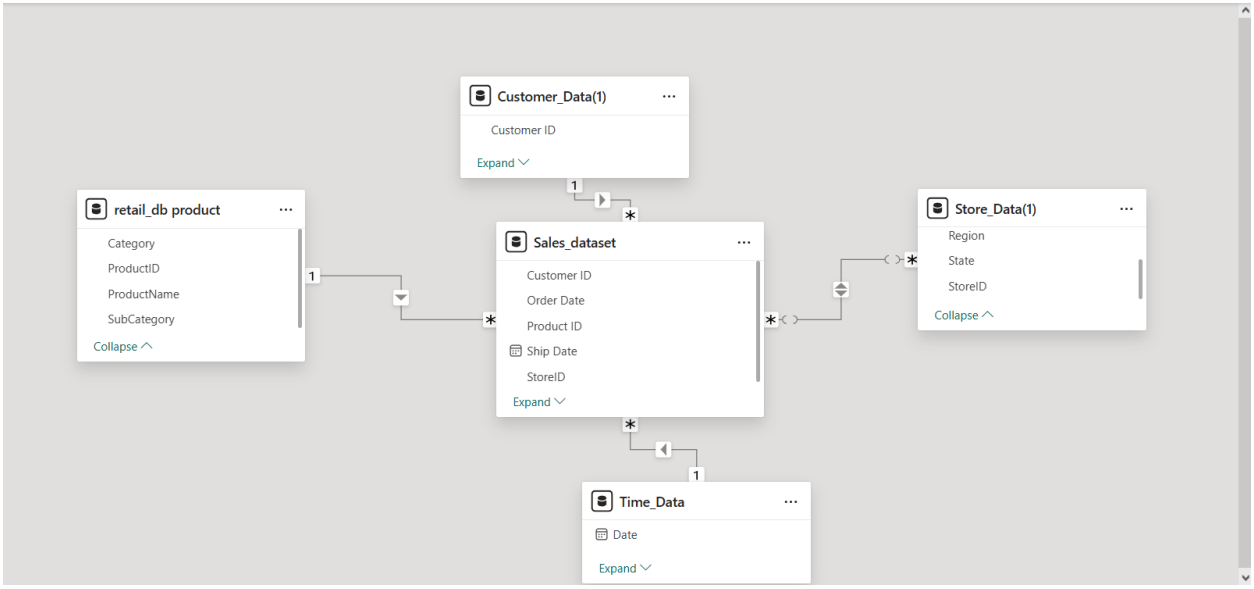

Retail_dataset_exp1 • Last saved: Today at 11:32 AM
Search
RISHAMADHURI K
Home
Open
Save
Save as
Share
Get data
Import
Export
Publish
Sign out
Options and settings
About
New
Blank report
Recommended
Getting started
Intro—What is Power BI?
Recent
Shared with me
Filter by keyword
Filter
Name
Location
Opened
hr employee.pbix
C:\Users\risha\OneDrive\Documents\Desktop\hr employee.pbix
21 hours ago
2.pbix
C:\Users\risha\OneDrive\Documents\Desktop\2.pbix
3 days ago

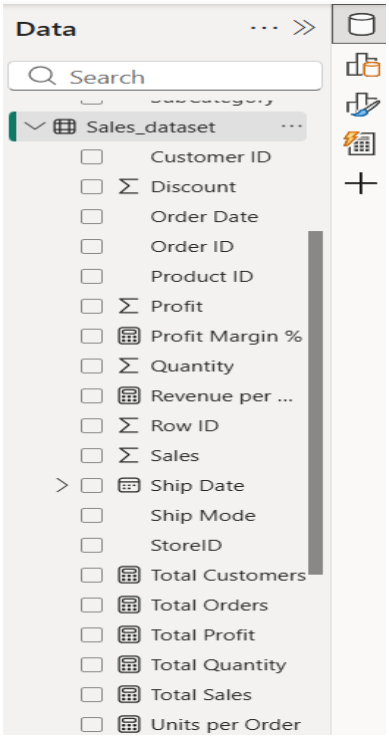

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
amitdb
companydb
finance_db
normalization
normalizationdb
parks_and_recreation
sakila
sys
Administration Schemas
Information
No object selected
Product_MySQL_Setup
-- Weekly Assignment 1 : Product master table for MySQL
-- Run this entire script in MySQL Workbench (or mysql CLI)
CREATE DATABASE IF NOT EXISTS retail_db;
USE retail_db;
DROP TABLE IF EXISTS Product;
CREATE TABLE Product (
Result Grid
ProductID ProductName Category SubCategory UnitCost
101 Wireless Mouse Electronics Accessories 300.00
102 Bluetooth Speaker Electronics Audio 900.00
103 Office Chair Furniture Seating 2500.00
104 Study Table Furniture Desks 3200.00
105 Running Shoes Footwear Sportswear 1400.00
106 Backpack Accessories Bags 800.00
107 LED Desk Lamp Electronics Lighting 450.00
Output
Action Output
2 00:47:44 USE retail_db 0 row(s) affected 0.000 sec
3 00:47:44 DROP TABLE IF EXISTS Product 0 row(s) affected, 1 warning(s): 1051 Unknown table 'retail_db.product' 0.015 sec
4 00:47:44 CREATE TABLE Product (ProductID INT PRIMARY KEY, ProductName VARCHAR(100), Category ... 0 row(s) affected 0.047 sec
5 00:47:44 INSERT INTO Product (ProductID, ProductName, Category, SubCategory, UnitCost) VALUES (101, 'Wireless ... 8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0 0.000 sec
6 00:47:44 SELECT * FROM Product LIMIT 0, 1000 8 row(s) returned 0.016 sec / 0.000 sec

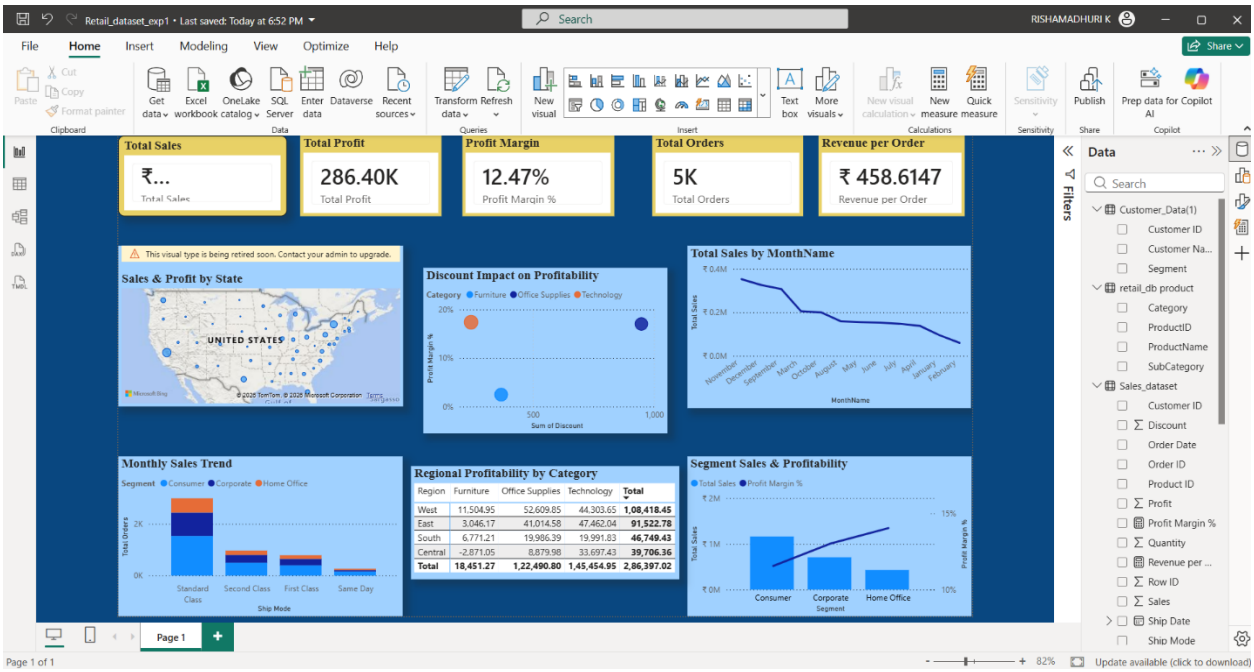

MySQL database
Server
127.0.0.1:3306
Database
retail_db
Advanced options
OK
Cancel


Untitled - Power Query Editor
Queries [5]
Customer_Data(1)
Store_Data(1)
Time_Data
Sales_dataset
retail_db product
Query Settings
PROPERTIES
Name
Sales_dataset
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
Changed Type


Retail_dataset_exp1 • Last saved: Today at 11:32 AM
Search
RISHAMADHURI K
File
Home
Help
Share
Properties
Cards
Data
Sales_dataset
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Quantity
Row ID
Sales
Collapse
retail_db product
Category
ProductID
ProductName
SubCategory
Collapse
Customer_Data(1)
Customer ID
Customer Name
Segment
Collapse
Store_Data(1)
Region
State
StoreID
Collapse
Time_Data
Date
Expand
All tables
Show the database in the header when applicable
No
Show related fields when card is collapsed
Yes
Pin related fields to top of card
No
100%
Update available (click to download)


Customer_Data(1)
Customer ID
Expand
retail_db product
Category
ProductID
ProductName
SubCategory
Collapse
Sales_dataset
Customer ID
Order Date
Product ID
Ship Date
StoreID
Expand
Store_Data(1)
Region
State
StoreID
Collapse
Time_Data
Date
Expand


Data
Search
Sales_dataset
Customer ID
Discount
Order Date
Order ID
Product ID
Profit
Profit Margin %
Quantity
Revenue per ...
Row ID
Sales
Ship Date
Ship Mode
StoreID
Total Customers
Total Orders
Total Profit
Total Quantity
Total Sales
Units per Order


Retail_dataset_exp1 • Last saved: Today at 6:52 PM
Search
RISHAMADHURI K
File
Home
Insert
Modeling
View
Optimize
Help
Share
Total Sales
₹...
Total Profit
286.40K
Total Profit
Profit Margin
12.47%
Profit Margin %
Total Orders
5K
Total Orders
Revenue per Order
₹ 458.6147
Revenue per Order
This visual type is being retired soon. Contact your admin to upgrade.
Sales & Profit by State
UNITED STATES
Discount Impact on Profitability
Total Sales by MonthName
Monthly Sales Trend
Regional Profitability by Category
Region Furniture Office Supplies Technology Total
West 11,504.95 52,609.85 44,303.65 1,08,418.45
East 3,046.17 41,014.58 47,462.04 91,522.78
South 6,771.21 19,986.39 19,991.83 46,749.43
Central -2,871.05 8,879.98 33,697.43 39,706.36
Total 18,451.27 1,22,490.80 1,45,454.95 2,86,397.02
Segment Sales & Profitability
Filters
Data
Search
Customer_Data(1)
retail_db product
Sales_dataset
Page 1
Page 1 of 1
Update available (click to download)

The final Power BI dashboard provides:

- Key metrics (Total Sales, Total Profit, Profit Margin %, Total Orders, Revenue per Order) through KPI cards.
- State-wise sales and profit performance on a map.
- Discount vs. profitability analysis by category.
- Monthly sales trends and ship-mode patterns.
- Regional profitability by category (Furniture, Office Supplies, Technology).
- Segment-wise sales and profit comparison (Consumer, Corporate, Home Office).

The dashboard gives management a single, centralized view to track performance and make faster, data-driven decisions.

Conclusion

This project connected Power BI to multiple data sources — a MySQL database and Excel/CSV flat files — and combined them into a single data model. Power Query was used for data cleaning and transformation, and relationships were built to link the tables. DAX measures were created to summarise key business metrics, and an interactive dashboard was built to show sales, profit, regional, and segment-level insights. This shows how Power BI can bring together data from different sources into one centralized reporting system for better business decisions.

Learning Outcomes

Through this project, the following was learned:

- How to connect Power BI to a MySQL database and Excel/CSV flat files using Get Data.
- Data cleaning and transformation techniques in Power Query, including data type conversion and header promotion.
- How to build a data model using one-to-many relationships between tables.
- How to write DAX measures for KPIs like Profit Margin % and Revenue per Order.
- How to design an interactive dashboard using KPI cards, maps, charts, and tables.
- How to turn raw multi-source data into useful business insights for reporting.

Business Insights

- Technology and Office Supplies have better profit margins than Furniture, even with lower total sales.
- West and East regions are the most profitable, while Central shows a loss in Furniture.
- Higher discounts lead to lower profit margins, especially in Furniture.
- Sales decline from November through mid-year, pointing to seasonal trends.
- Consumer segment has the highest order volume, while Corporate shows better profit margins.